

Multiple Choice (10 marks)

1. Which among the following is the least strong security encryption standard?
 - (a) WEP
 - (b) WPA
 - (c) WPA 2
 - (d) WPA 3
2. _____ is also a part of darknet that is employed for transferring files anonymously.
 - (a) Freenet
 - (b) ARPANET
 - (c) Stuxnet
 - (d) Internet
3. When does a buffer overflow occur, generally speaking?
 - (a) when writing to a pointer that has been freed
 - (b) when copying a buffer from the stack to the heap
 - (c) when a pointer is used to access memory not allocated to it
 - (d) when the program notices a buffer has filled up, and so starts to reject requests
4. Public key encryption is advantageous over Symmetric key Cryptography because of
 - (a) Speed
 - (b) Space
 - (c) Key exchange
 - (d) Key length
5. The _____ can cost you money, by sending text messages from your mobile phone numbers.
 - (a) IM - Trojans
 - (b) Backdoor Trojans
 - (c) SMS Trojan
 - (d) Ransom Trojan

True / False (10 marks)

Answer True or False

6. True or False: A MAC can be secure even if its tag length is only 5 bits, due to strong underlying cryptographic methods.
7. True or False: WPA relies on LEAP for encryption, which is considered a secure method for wireless communication.
8. True or False: When the STP solver times out, EXE assumes the query is satisfiable and continues executing the path.
9. True or False: Mutation-based fuzzing works by making small mutations to the target program itself rather than the input data.
10. True or False: Blackbox fuzzers can effectively cover every line of code since they analyze program outputs without internal insights.

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

11. Write a function to count repeated items of a tuple.
12. Write a python function to remove spaces from a given string.

End of Paper